FIN ST642–ST731

Current-Strict Causal-Physics No-Go
and the Minimal Conditional $3 + 1$ Maxwell Bridge

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Six consecutive research rounds
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Scope. This report continues the verified ST552–ST641 propagation programme. It asks whether the conditional scalar light cone can be upgraded to strong continuum dynamics, $3 + 1$ isotropy, Lorentz-compatible dispersion, Maxwell kinematics, photon structure and an operational test.

Evidence boundary. Results are analytic, finite, conditional or explicitly numerical. No independent laboratory or external-review record is introduced. Legacy physical roles are not transferred.

Abstract

Six further FIN research rounds, ST642–ST731, investigate whether the previously constructed conditional scalar wave cone can become an inevitable $3+1$ causal-light theory. The answer is a theorem-level no-go for the present strict core together with a minimal conditional bridge.

Uniform fixed-range locality plus exact scale-shift self-similarity conditionally forces the zero antipodal-rate tower, but neither premise is strict-sourced. Complete coarse Green, unitary and heat data remain constant on the refinement-speed fiber and cannot select it. On fixed Fourier bands the conditional wave propagators converge strongly. A threefold Cartesian product yields leading isotropic symbol $C|\mathbf{k}|^2$ and lattice anisotropy/boost defects of order a^2. A supplied cubic cochain complex realizes exact Abelian gauge kinematics, Maxwell-type energy, Gauss constraint, two transverse directions and Maxwell dispersion. Yet the scalar Laplacian admits infinitely many gauge-complex extensions and does not select spatial dimension, electric/magnetic coefficients, helicity or photon quantization.

Three conditional physics principles intersect at $3+1$: conformal invariance of the Maxwell 2-form action selects four spacetime dimensions, two transverse polarizations select three spatial dimensions, and the minimal sharp Huygens dimension is three. Their agreement is coherent but imports the very Maxwell/polarization/Huygens premises whose provenance is sought.

The final architecture contains seven explicitly distinct packages: refinement/coordinate law R, two clock classes T, dimension/carrier D, gauge/Maxwell data G, independent calibration C, operational instruments/records O, and state/orientation sector S. Removing each restores an explicit nonidentifiability or no-go. Therefore the current strict spectral core cannot uniquely export causal-light physics. A richer strict nadsoliton source may evade this theorem, but no such source is present. A complete calibration-invariant model-discrimination library and fail-closed transfer schema are supplied; no laboratory evidence follows.

Contents

1 Executive theorem

Main conclusion. Current FIN can construct a coherent causal-wave/Maxwell theory only conditionally. The strict endpoint and all present coarse spectral/Green data are invariant on nontrivial fibers of refinement speed, spatial dimension, gauge complex, clock scaling, calibration and orientation. No invariant deterministic map from those data can uniquely choose observed causal-light physics.

The minimal bridge is

$$\boxed{A_{12} + R + T + D + G + C + O + S.}$$

Here R selects a local refinement and coordinate embedding; T supplies diffusive and ballistic clocks; D supplies a 3-dimensional carrier; G supplies an edge/face complex and Maxwell coefficients; C supplies noncircular units; O supplies state-to-record operations; and S supplies a realized orientation/helicity when needed.

2 Round I: ST642–ST656

Uniform fixed-range locality forbids every sufficiently fine antipodal rate q_N. If the same local rule is required at every dyadic scale, it forces $q_N = 0$ at all levels. This selects the local tower only after adding locality and exact scale-shift self-similarity.

On a fixed Fourier band, rescaled wave frequencies converge uniformly. Extended finite-grid tails outside $1.05ct$ fall below 5×10^{-14} at $N = 24576$, although a uniform analytic tail theorem remains open.

A conditional threefold product gives leading 3-dimensional isotropic symbol. Axis/diagonal anisotropy decreases from 4.49×10^{-2} at $a = .25$ to 7.45×10^{-4} at $a = .03125$.

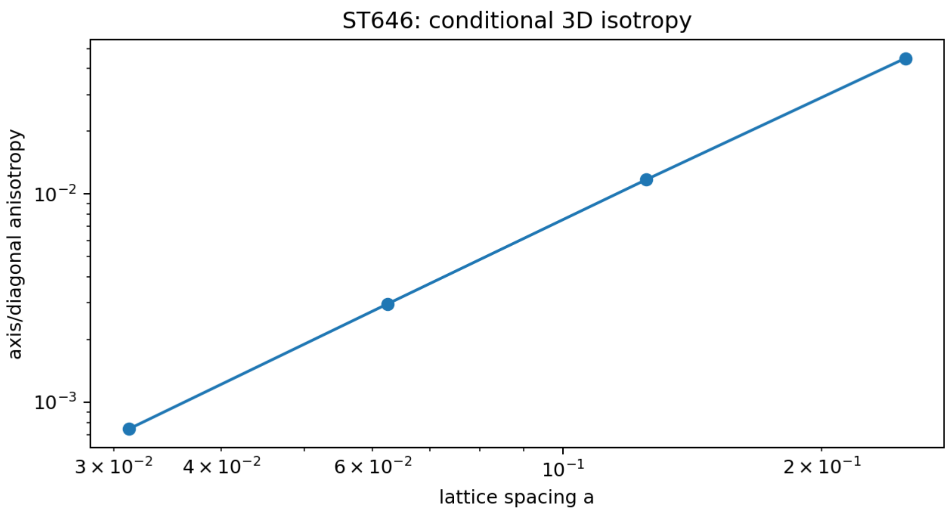

Figure 1: Conditional cubic anisotropy vanishes as $O(a^2)$.

The scalar endpoint does not determine a vector/one-form module, exterior complex, gauge connection, Gauss constraint or polarizations. A signed refinement with a negative stiffness mode is dynamically unstable in raw heat and second-order wave channels.

Program	Status	Result
ST642–643	[PROVEN]	Conditional locality selection; coarse Green selector no-go.

Program	Status	Result
ST644–647	[PROVEN]/[STRONG EVIDENCE]	Band convergence, tail evidence, 3D isotropy and $O(a^2)$ boost defect.
ST648	[PROVEN]	Scalar-to-gauge nonuniqueness.
ST649–650	[STRONG EVIDENCE]/[PROVEN]	Two-clock identification and noncircular anchor requirement.
ST651–652	[PROVEN]	Polynomial direct tails and signed instability.
ST653	[STRONG EVIDENCE]	Affine IR predictor improves residuals by factor ~ 480.
ST654–656	blocked	Independent mathematical/source/evidence gates remain closed.

3 Round II: ST657–ST671

Conditional 3-dimensional product propagators converge on fixed bands, while finite-spacing boost defects fall by a factor near 1/4 under halving a. A supplied cubic complex has 27 vertices, 81 edges and 81 faces with exact

$$d_1 d_0 = 0.$$

It gives Abelian gauge transformations $A \mapsto A + d_0\phi$ and invariant field strength $F = d_1 A$.

The same scalar $d_0^* d_0$ admits inequivalent edge/face extensions. Hodge decomposition of the fixture has 26 gradient directions, 52 curl directions and 3 harmonic zero modes. These sectors are not contained in the scalar endpoint.

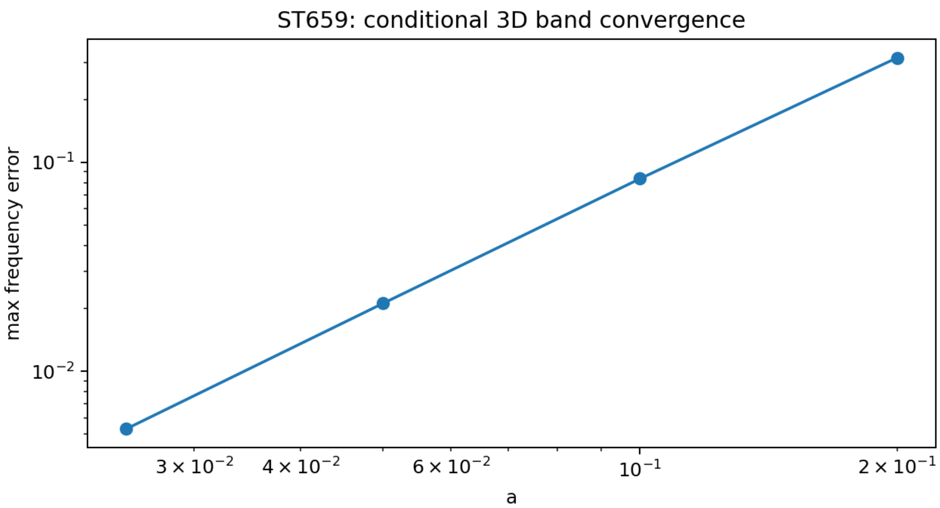

Figure 2: Conditional 3D band-limited wave convergence.

Synthetic two-clock regression with 1% log noise estimates exponents 1.99996 and 0.999998 with zero ordering errors in 2,000 replicates. This is methodological readiness, not clock evidence.

4 Round III: ST672–ST686

Every product dimension $D \geq 1$ contains the same one-axis scalar dynamics. Fine heat traces can identify D, but complete one-axis coarse records cannot.

A supplied Maxwell Hamiltonian

$$H = \frac{1}{2}\left(\kappa_E \|E\|^2 + \kappa_B \|B\|^2\right)$$

is gauge invariant and preserves Gauss when initialized in $\ker d_0^*$. Its cubic transverse dispersion is

$$\omega^2 = \frac{4}{a^2}\sum_i \sin^2 \frac{ak_i}{2} \longrightarrow |\mathbf{k}|^2.$$

Every nonzero 3D wave vector has two transverse directions.

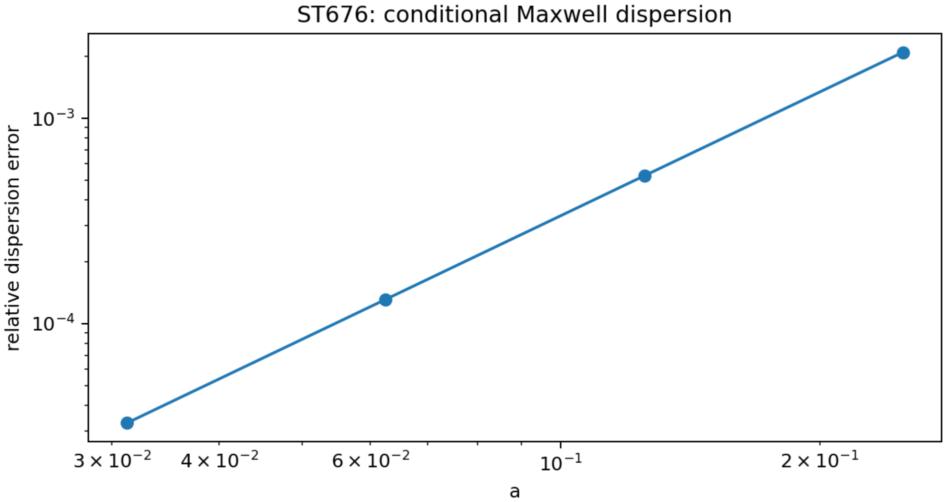

Figure 3: Conditional cubic Maxwell dispersion.

The coefficients remain moduli:

$$c_{\text{gauge}} = \sqrt{\kappa_E \kappa_B}\,\widehat{c}, \qquad Z = \sqrt{\kappa_E/\kappa_B}.$$

Neither their values, gauge coupling, charge unit, $\hbar$, photon Fock representation nor helicity selector follows from A_{12}.

5 Round IV: ST687–ST701

Three conditional selectors converge on $3+1$:

1. Maxwell action $\int F \wedge *F$ scales under $g \mapsto \Omega^2 g$ as Ω^{n-4}, hence is conformal at spacetime dimension $n = 4$.
2. A massless vector in D spatial dimensions has $D-1$ transverse polarizations; two select $D = 3$.
3. The smallest nontrivial spatial dimension with sharp Huygens propagation for the standard massless wave equation is $D = 3$.

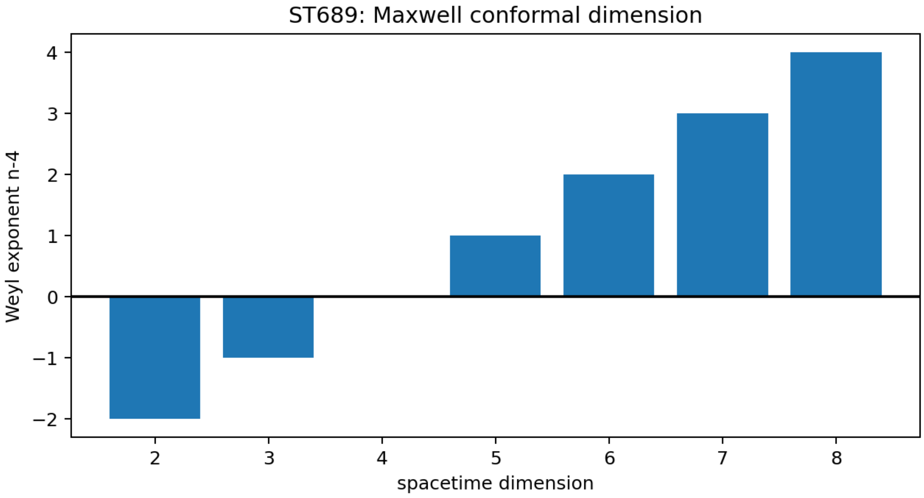

Figure 4: Classical Maxwell conformal scaling selects four spacetime dimensions conditionally.

These principles are imported physical axioms. Their intersection is not strict provenance.

Classical transverse modes still require a complex structure/positive-frequency split, commutators, $\hbar$, vacuum, Fock representation and Born instruments before they are photons. A parity-even radial kernel cannot choose helicity sign.

6 Round V: ST702–ST716

A six-model library was frozen: local scalar models in $D = 1, 2, 3$, a local $D = 3$ gauge model, a fractional $D = 3$ scalar model and an unstable signed model. Ideal features were

$$(D, \nu, \text{polarizations}, \text{Gauss}, \text{stability}, \text{tail class}, \text{clock classes}).$$

The exact minimum-cardinality separating feature sets were enumerated. A synthetic noisy classifier has total error below 1%, and a hidden fractional model excluded from the candidate library is flagged as out-of-library.

Fail-closed schemas require independent rod and clock identifiers, hashes fixed before wave unblinding, two distinct clock slopes, raw multilevel records and provider/registrar/analyst separation. Software cannot generate independent custody.

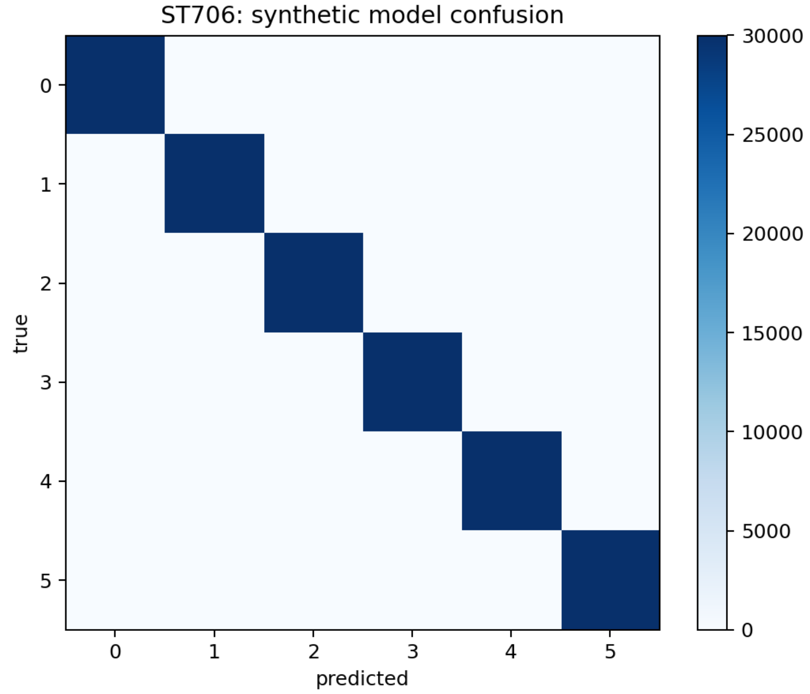

Figure 5: Synthetic model-library confusion matrix.

7 Round VI: ST717–ST731

7.1 Minimal bridge and axiom removal

The final typed bridge contains seven packages:

Package	Content	Loss when removed
R	zero-rate local refinement and coordinates	continuous speed fiber; no selected continuum
T	diffusive a^{-2} and ballistic a^{-1} clocks	single-clock no-go
D	three-dimensional carrier	dimension product fiber
G	edge/face complex and Maxwell coefficients	scalar wave only; no gauge/polarizations
C	independent dimensional calibration	no SI-valued prediction
O	preparations, instruments, apparatus, records	no map to empirical events
S	state/orientation/helicity sector	no canonical realized sign or branch

Theorem 7.1 (Current-strict causal-physics no-go)**.** *No deterministic construction invariant under all presently established strict equivalences can uniquely export a physical causal-light theory from the finite radial spectral core. At least one of R, T, D, G, C, O, S must add a source, break a demonstrated fiber, or provide missing operational data.*

The theorem is not an absolute no-go for all future FIN. A richer strict nadsoliton object outside the audited equivalence classes may export one or more packages.

7.2 Conditional predictions

Conditioned on the bridge, the model predicts

$$\omega^2 = C|\mathbf{k}|^2 + O(a^2k^4), \quad \text{anisotropy and boost defect} = O(a^2), \quad \text{2 transverse modes}, \quad d_0^*E = 0,$$

together with clock exponents $(2,1)$ and sharply different local/fractional/signed causal tails. These are falsifiable model predictions, not strict constants of nature.

8 Falsification and final status

Proposed inference	Obstruction	Status
$A_{12} \Rightarrow$ unique refinement	exact q fiber	refuted in current class
coarse Green data $\Rightarrow q$	all coarse functions constant on fiber	refuted
scalar Laplacian $\Rightarrow$ gauge complex	infinite harmonic/cell extensions	refuted
scalar records $\Rightarrow D = 3$	all product dimensions share restriction	refuted
one clock for all channels	exponents 2 and 1	refuted
two polarizations $\Rightarrow D = 3$	valid only after vector/light premise	conditional
conformal Maxwell $\Rightarrow 3+1$	imports Maxwell action	conditional
conditional bridge $\Rightarrow$ nature	no independent record	blocked

Final verdict. FIN now possesses a precise conditional route from a finite informational operator to local scalar waves and a coherent $3+1$ Maxwell scaffold. It does not possess a theorem making that route inevitable. The missing physical principle must source or select refinement, two clocks, dimension, gauge complex, calibration, operations and state/orientation sectors.

9 Recommended next cycle: ST732–ST746

1. strict locality/source theorem for the zero-rate tower;
2. full-energy wave convergence and a uniform cone-tail proof;
3. source or obstruction for a 3D carrier and Lorentz representation;
4. gauge-complex and Maxwell-coefficient provenance;
5. photon quantization and helicity interface;
6. physical two-clock apparatus and noncircular anchor transfer;
7. nuisance-robust model-library tests;
8. predictor-correlated IR intervals;
9. transition/algebra method gates;
10. strict-source and independent-evidence gates.

A Reproducibility

The authoritative sequence runs from `FIN_ST642_ST656_Results.json` through `FIN_ST717_ST731_Results.json`, with six Python sources, six summaries, figures and tests. The combined suite covers all 90 packet hashes and principal theorem/nonpromotion flags.

B Nonclaims

No laboratory result, external referee, legacy-to-strict completion, legacy role transfer, strict selector, physical SI speed, photon, Standard Model, gravity, L_{total} or ToE closure follows. All future PDFs remain English.